Operating System Exam (2021/11)

1. (5 pt) Ordinarily the `exec()` system call follows the `fork()`. Explain what would happen if a programmer were to inadvertently place the call to `exec()` before the call to `fork()`.

2. (5 pt) Why does the socket communication use the port number instead of process id number to identify server or client process?

3. (8 pt) What is processor affinity? Explain why it needed to be considered during process scheduling on multiple CPU systems.

4. (7 pt) Explain why implementing synchronization primitives by disabling interrupts is not appropriate in a single-processor system if the synchronization primitives are to be used in user-level programs.

5. (15 pt) Discuss how the following pairs of scheduling criteria conflict in certain settings.

   5.1 CPU utilization and response time
   5.2 Average turnaround time and maximum waiting time
   5.3 I/O device utilization and CPU utilization
   5.4 Deadline and fairness
   5.5 Throughput and context switch overhead

6. Multilevel queue scheduling

   6.1 (5 pt) List two possible scheduling among queue, which we discussed in class.

   6.2 (10 pt) Modify the following multilevel queue scheduling system such that a new process will automatically enter the appropriate queue after a few runs.

system process
faculty interactive process
student interactive process
faculty batch process
student batch process

7. Multilevel queue scheduling

7.1 (5 pt) What is **busy-waiting** in the discussion of critical section problem? Explain how **busy-waiting** may degrade the system performance.

7.2 (10 pt) How to implement semaphore without **busy-waiting** using `block` and `wakeup`? You need to write down the definition of data structure `semaphore` and `wait` and `signal` operations.

8. (10 pt) We have discussed the correct solution to the consumer-producer problem, which does not use any synchronization mechanism, as shown below. Please explain why it can only use `BUFFER_SIZE-1` elements of buffer.

Producer:

```
while (true){
        /* produce an item in next produced */
        while (((in + 1) % BUFFER_SIZE) == out)
                ; /*do nothing */
        buffer[in] = next_produced;
        in = (in + 1) % BUFFER_SIZE;
}
```

Consumer:

```
while (true){
        while (in == out)
                ; /* do nothing */
        next_consumed = buffer[out];
        out = (out + 1) % BUFFER_SIZE;

        /* consume the item in next consumed */
}
```

9. Consider two processes, **P_1** and **P_2**, where $p_1 = d_1 = 50$, $t_1 = 25$, $p_2 = d_2 = 75$, and $t_2 = 30$.

9.1 (10 pt) Can these two processes be scheduled using rate-monotonic scheduling (RMS)? Illustrate your answer using a Gantt chart such as the ones we did in class.

9.2 (10 pt) Illustrate the scheduling of these two processes using earliest-deadline-first (EDF) scheduling.